

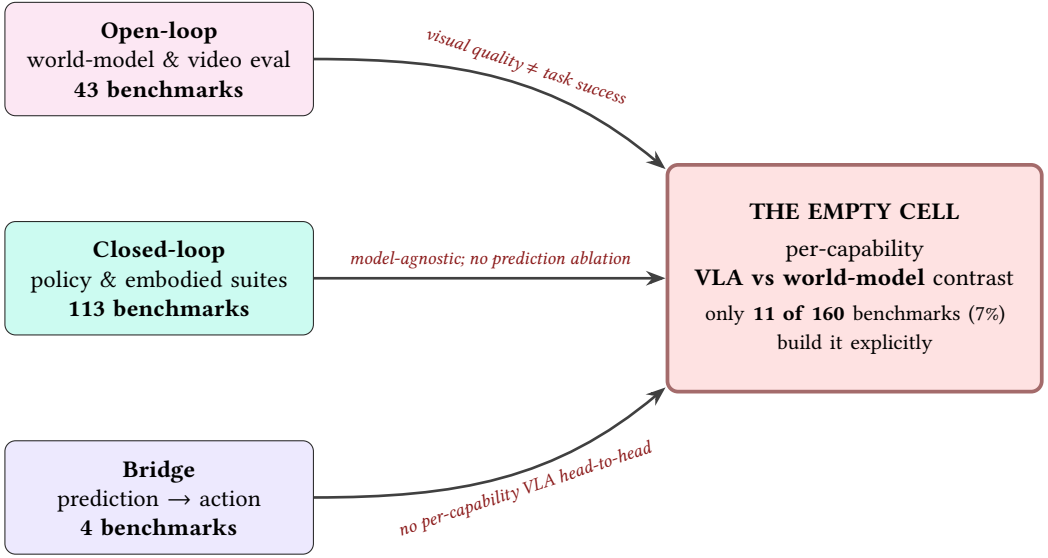


Fig. 1. **The question no benchmark answers.** The robot-evaluation landscape splits into three lanes: *open-loop* suites (43) score a world model’s prediction or generation quality but never execute it; *closed-loop* suites (113) score task success but are model-agnostic and never build a direct Vision-Language-Action vs world-model contrast; and a handful of *bridges* (4) reach executed action yet omit the per-capability head-to-head. The empty cell each lane points to, a per-capability contrast that isolates whether prediction earns a closed-loop advantage, is built explicitly by only **11 of 160** benchmarks (7%). This survey maps the landscape by that missing axis (Table ??).

REFERENCES

- [1] Pranav Atreya, Karl Pertsch, Tony Lee, Moo Jin Kim, Arhan Jain, Artur Kuramshin, Clemens Eppner, Cyrus Neary, Edward Hu, Fabio Ramos, Jonathan Tremblay, Kanav Arora, Kirsty Ellis, Luca Macesanu, Marcel Torne Villasevil, Matthew Leonard, Meedeum Cho, Ozgur Aslan, Shivin Dass, Jie Wang, William Reger, Xingfang Yuan, Xuning Yang, Abhishek Gupta, Dinesh Jayaraman, Glen Berseth, Kostas Daniilidis, Roberto Martin-Martin, Youngwoon Lee, Percy Liang, Chelsea Finn, and Sergey Levine. 2025. RoboArena: Distributed Real-World Evaluation of Generalist Robot Policies. arXiv:2506.18123.
- [2] Florian Bordes, Quentin Garrido, Justine T Kao, Adina Williams, Michael Rabbat, and Emmanuel Dupoux. 2025. IntPhys 2: Benchmarking Intuitive Physics Understanding In Complex Synthetic Environments. arXiv:2506.09849.
- [3] Yu-Wei Chao, Chris Paxton, Yu Xiang, Wei Yang, Balakumar Sundaralingam, Tao Chen, Adithyavairavan Murali, Maya Cakmak, and Dieter Fox. 2022. HandoverSim: A Simulation Framework and Benchmark for Human-to-Robot Object Handovers. arXiv:2205.09747.
- [4] Changan Chen, Yuejiang Liu, Sven Kreiss, and Alexandre Alahi. 2018. Crowd-Robot Interaction: Crowd-aware Robot Navigation with Attention-based Deep Reinforcement Learning. arXiv:1809.08835.
- [5] Delong Chen, Willy Chung, Yejin Bang, Ziwei Ji, and Pascale Fung. 2025. WorldPrediction: A Benchmark for High-level World Modeling and Long-horizon Procedural Planning. arXiv:2506.04363.
- [6] Nikita Chernyadev, Nicholas Backshall, Xiao Ma, Yunfan Lu, Younggyo Seo, and Stephen James. 2024. BiGym: A Demo-Driven Mobile Bi-Manual Manipulation Benchmark. arXiv:2407.07788.
- [7] Xiaowei Chi, Chun-Kai Fan, Hengyuan Zhang, Xingqun Qi, Rongyu Zhang, Anthony Chen, Chi min Chan, Wei Xue, Qifeng Liu, Shanghang Zhang, and Yike Guo. 2024. EVA: An Embodied World Model for Future Video Anticipation. arXiv:2410.15461.
- [8] Daniel Dauner, Marcel Hallgarten, Tianyu Li, Xinshuo Weng, Zhiyu Huang, Zetong Yang, Hongyang Li, Igor Gilitschenki, Boris Ivanovic, Marco Pavone, Andreas Geiger, and Kashyap Chitta. 2024. NAVSIM: Data-Driven Non-Reactive Autonomous Vehicle Simulation and Benchmarking. arXiv:2406.15349.

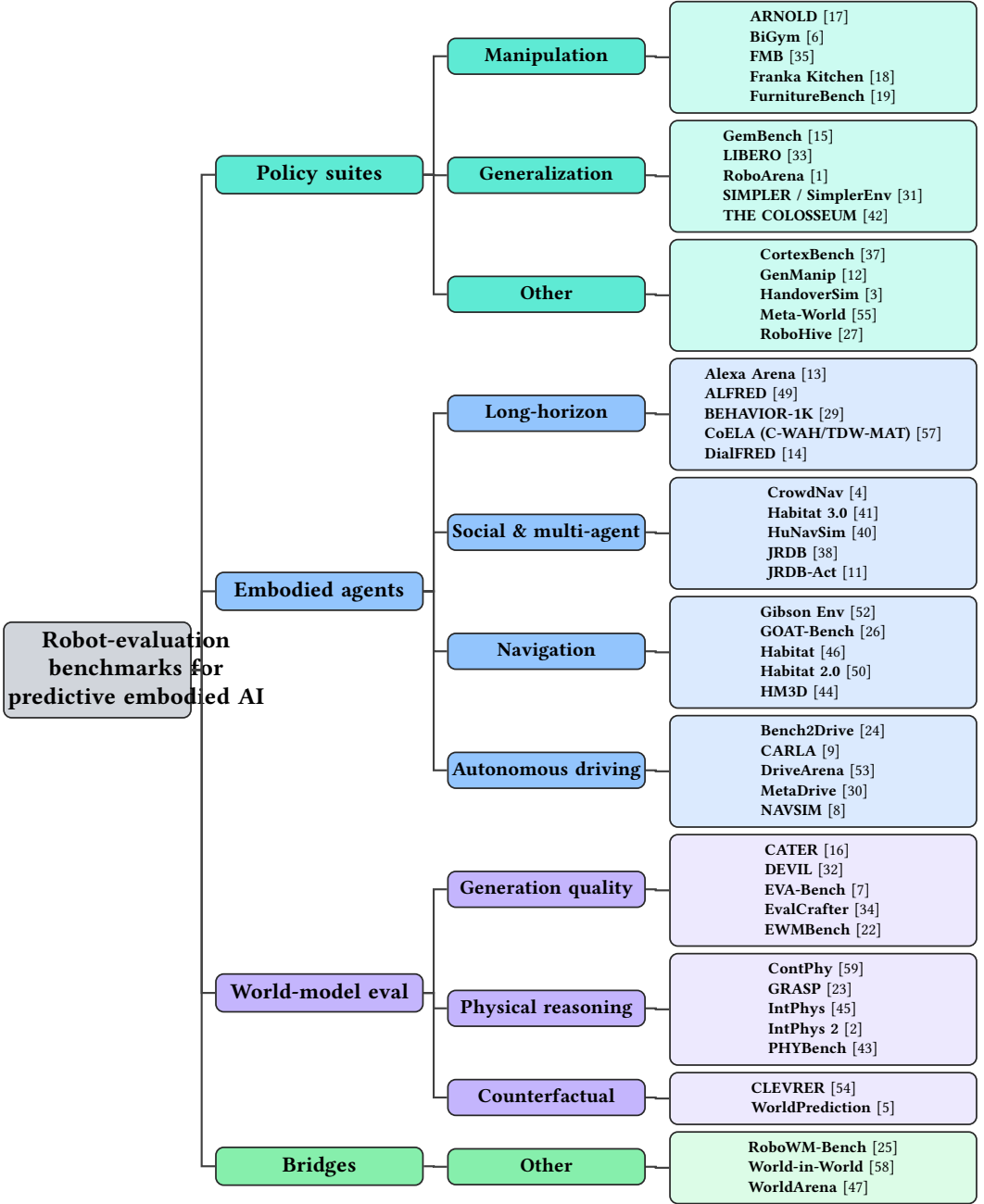


Fig. 2. **A taxonomy of robot-evaluation benchmarks for predictive embodied intelligence** (50 representative benchmarks of the 160 in Table ??). Benchmarks are grouped by evaluation lane and then by the capability they stress; each line names a benchmark and what it measures. The lanes partition the field by *how* a benchmark scores a policy (open-loop prediction quality, closed-loop task success, or a prediction-to-action bridge); the capability clusters are the axis β along which the survey asks whether prediction earns a closed-loop advantage.

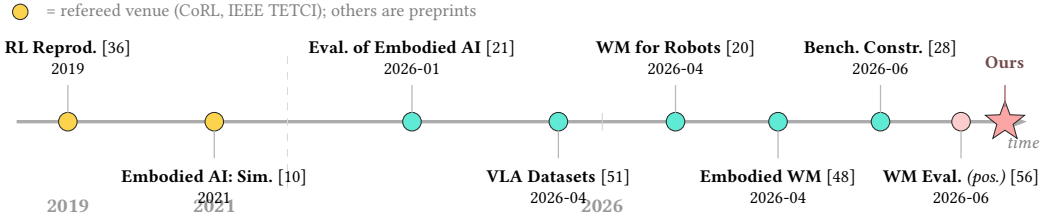


Fig. 3. The closest benchmark/evaluation surveys as a timeline. Two predate the recent wave, an IEEE TETCI survey of embodied-AI simulators and tasks [10] and a CoRL survey of real-robot RL reproducibility [36] (amber = refereed venue); the rest are 2026 preprints on embodied evaluation, VLA data, and world models. None organises the robot-evaluation landscape by evaluation-mode $\times$ capability $\times$ model-family; the sole work reaching the advantage-of-prediction axis is a *position* paper [56]. **Ours** (★) fills that cell; per-axis coverage is in Table ??.

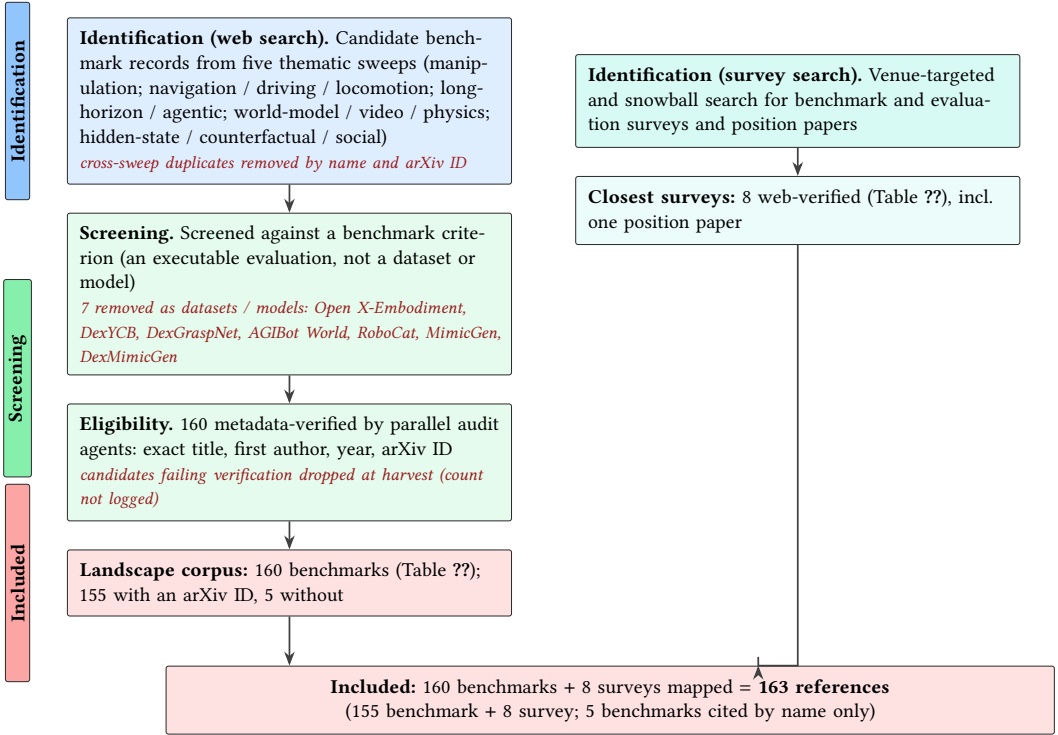


Fig. 4. **Corpus construction** in the PRISMA 2020 style [39]. The left column is the web-search stream behind the landscape corpus (Table ??): five thematic sweeps, de-duplicated by name and arXiv ID, screened to drop datasets and models, and metadata-verified by parallel audit agents. The right column is the venue-targeted and snowball search behind the closest surveys (Table ??). Both streams feed the included set. Candidates whose metadata could not be verified were discarded at harvest; their count was not logged, which we state rather than estimate.

[9] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. 2017. CARLA: An Open Urban Driving Simulator. arXiv:1711.03938.

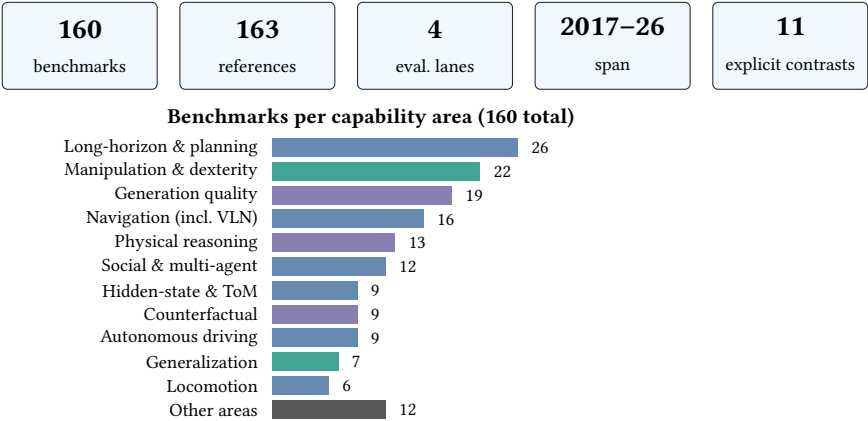


Fig. 5. **The catalogued corpus at a glance.** The survey maps 160 web-verified benchmarks across 4 evaluation lanes, spanning 2017–2026 and totalling 163 references; only 11 build an explicit Vision-Language-Action vs world-model contrast. The bars give benchmarks per capability area, with kindred clusters merged (dexterity into manipulation, vision-language navigation into navigation, theory-of-mind into hidden-state); the twelve areas sum to the 160 works of Table ???. Bars are coloured by the dominant lane of each area (teal policy, blue embodied, purple world-model).

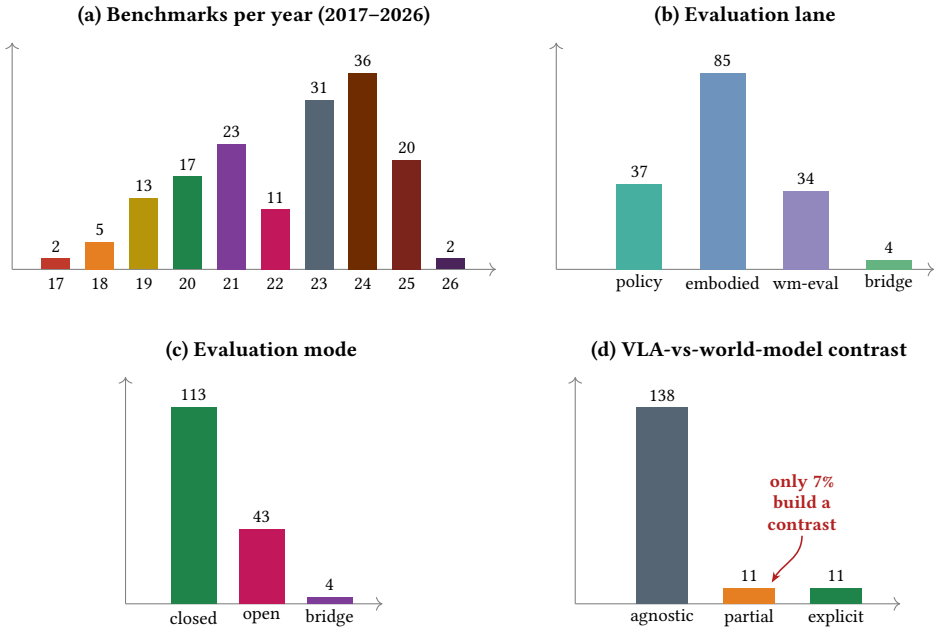


Fig. 6. **Profile of the 160 catalogue benchmarks** (Table ??), re-counted directly from the catalogue. (a) publication year, showing the 2023–2024 surge; (b) evaluation lane (embodied navigation dominates, then policy suites); (c) evaluation mode (closed-loop task success dominates, open-loop prediction is second, only four bridges reach action); (d) whether the benchmark builds a native Vision-Language-Action vs world-model contrast, the survey’s organising question: **138 of 160 are model-agnostic**, and only **11 (7%)** build the contrast explicitly. Counts are the exact column tallies of Table ??.

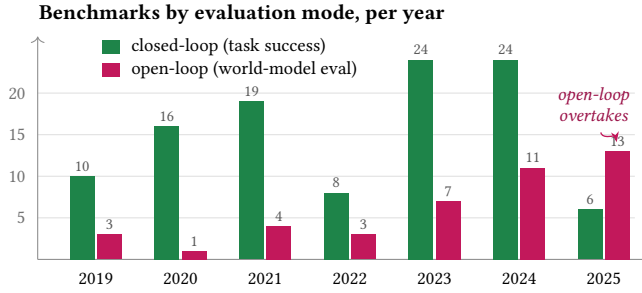


Fig. 7. **The world-model-evaluation wave is recent.** Per-year counts of closed-loop task-success benchmarks (green) and open-loop world-model / video-generation benchmarks (magenta) in the catalogue. Closed-loop suites dominate through 2024; in **2025 open-loop evaluation overtakes them** (13 vs 6) as the world-model-generation literature arrives. The four prediction-to-action bridges are newer still, all appearing in 2024 or later. Years are the benchmarks’ first-release years (2026 is partial and omitted); counts are the tallies of Table ??.

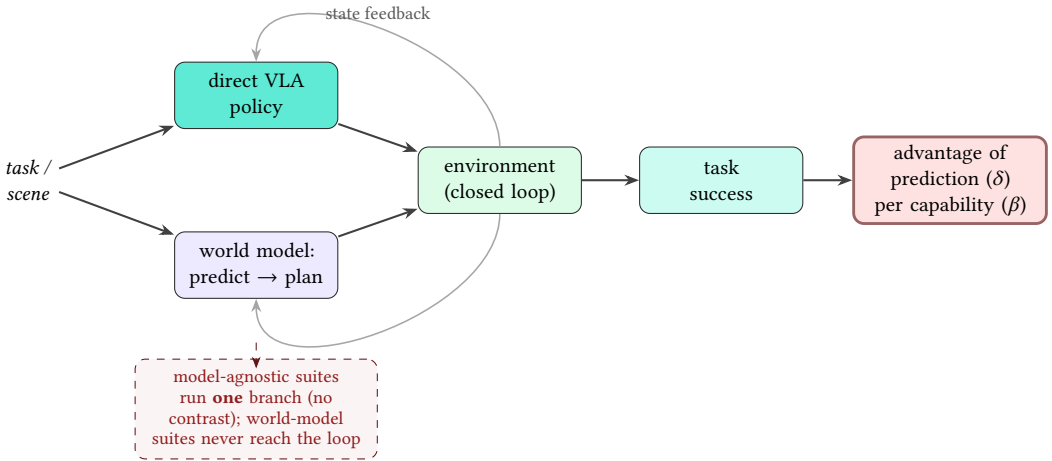


Fig. 8. **The evaluation loop that isolates the advantage of prediction.** A benchmark that can answer “does world modelling help a robot act” must run a direct Vision-Language-Action policy *and* a world-model policy (predict then plan) under one closed loop, then report the closed-loop gain (δ) sliced by capability (β). Today’s suites break this loop in one of two ways: model-agnostic policy suites host a single policy and never build the contrast, while world-model suites score the prediction open-loop and never execute it (§??).

- [10] Jiafei Duan, Samson Yu, Hui Li Tan, Hongyuan Zhu, and Cheston Tan. 2022. A Survey of Embodied AI: From Simulators to Research Tasks. *IEEE Transactions on Emerging Topics in Computational Intelligence* (2022). arXiv:2103.04918.
- [11] Mahsa Ehsanpour, Fatemeh Saleh, Silvio Savarese, Ian Reid, and Hamid Reza Tofighi. 2021. JRDB-Act: A Large-scale Dataset for Spatio-temporal Action, Social Group and Activity Detection. arXiv:2106.08827.
- [12] Ning Gao, Yilun Chen, Shuai Yang, Xinyi Chen, Yang Tian, Hao Li, Haifeng Huang, Hanqing Wang, Tai Wang, and Jiangmiao Pang. 2025. GENMANIP: LLM-driven Simulation for Generalizable Instruction-Following Manipulation. arXiv:2506.10966.
- [13] Qiaozi Gao, Govind Thattai, Suhaila Shakiah, Xiaofeng Gao, Shreyas Pansare, Vasu Sharma, Gaurav Sukhatme, Hangjie Shi, Bofei Yang, Desheng Zheng, Lucy Hu, Karthika Arumugam, Shui Hu, Matthew Wen, Dinakar Guthy, Cadence Chung, Rohan Khanna, Osman Ipek, Leslie Ball, Kate Bland, Heather Rocker, Yadunandana Rao, Michael Johnston, Reza

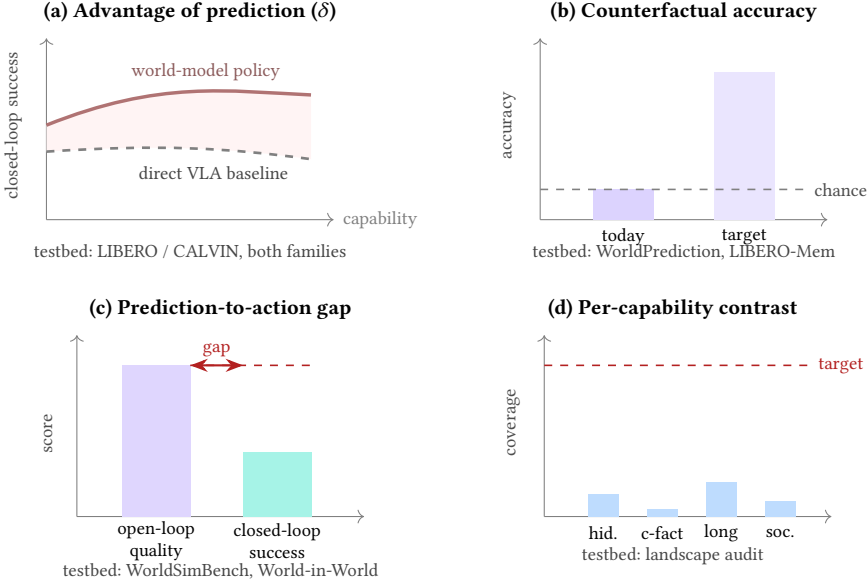


Fig. 9. **An actionable evaluation protocol for the future agenda** (Table ??). The panels are *schematic*: they show the axes and the *target* shape of each measurement, not measured results. (a) a world-model policy should beat a matched direct-VLA baseline in closed-loop success, and the per-capability gap is the advantage of prediction (δ), on LIBERO / CALVIN-style suites run with both families; (b) counterfactual accuracy is near chance today and should rise, on WorldPrediction and occlusion splits; (c) a world model’s open-loop quality far exceeds its closed-loop task success, and that gap must be reported, on bridge suites; (d) the fraction of capabilities with a native VLA-vs-world-model contrast is far below target across the landscape.

- Ghanadan, Arindam Mandal, Dilek Hakkani Tur, and Prem Natarajan. 2023. Alexa Arena: A User-Centric Interactive Platform for Embodied AI. arXiv:2303.01586.
- [14] Xiaofeng Gao, Qiaozi Gao, Ran Gong, Kaixiang Lin, Govind Thattai, and Gaurav S. Sukhatme. 2022. DialFRED: Dialogue-Enabled Agents for Embodied Instruction Following. arXiv:2202.13330.
- [15] Ricardo Garcia, Shizhe Chen, and Cordelia Schmid. 2024. Towards Generalizable Vision-Language Robotic Manipulation: A Benchmark and LLM-guided 3D Policy. arXiv:2410.01345.
- [16] Rohit Girdhar and Deva Ramanan. 2019. CATER: A diagnostic dataset for Compositional Actions and TEmporal Reasoning. arXiv:1910.04744.
- [17] Ran Gong, Jiangyong Huang, Yizhou Zhao, Haoran Geng, Xiaofeng Gao, Qingyang Wu, Wensi Ai, Ziheng Zhou, Demetri Terzopoulos, Song-Chun Zhu, Baoxiong Jia, and Siyuan Huang. 2023. ARNOLD: A Benchmark for Language-Grounded Task Learning With Continuous States in Realistic 3D Scenes. arXiv:2304.04321.
- [18] Abhishek Gupta, Vikash Kumar, Corey Lynch, Sergey Levine, and Karol Hausman. 2019. Relay Policy Learning: Solving Long-Horizon Tasks via Imitation and Reinforcement Learning. arXiv:1910.11956.
- [19] Minh Heo, Youngwoon Lee, Doohyun Lee, and Joseph J. Lim. 2023. FurnitureBench: Reproducible Real-World Benchmark for Long-Horizon Complex Manipulation. arXiv:2305.12821.
- [20] Bohan Hou, Gen Li, Jindou Jia, Tuo An, Xinying Guo, Sicong Leng, Haoran Geng, Yanjie Ze, Tatsuya Harada, Philip Torr, Oier Mees, Marc Pollefeys, Zhuang Liu, Jiajun Wu, Pieter Abbeel, Jitendra Malik, Yilun Du, and Jianfei Yang. 2026. World Model for Robot Learning: A Comprehensive Survey. arXiv:2605.00080.
- [21] Liyu Hou, Linyuan Gao, Yuan Wu, and Yi Chang. 2026. A Survey on Evaluation of Embodied AI. Authorea preprint, DOI:10.22541/au.176851544.45077723.
- [22] Yue Hu, Siyuan Huang, Yue Liao, Shengcong Chen, Pengfei Zhou, Liliang Chen, Maoqing Yao, and Guanghui Ren. 2025. EWM-Bench: Evaluating Scene, Motion, and Semantic Quality in Embodied World Models. arXiv:2505.09694.
- [23] Serwan Jassim, Mario Holubar, Annika Richter, Cornelius Wolff, Xenia Ohmer, and Elia Bruni. 2023. GRASP: A novel benchmark for evaluating language GROUNDING And Situated Physics understanding in multimodal language models.

- arXiv:2311.09048.
- [24] Xiaosong Jia, Zhenjie Yang, Qifeng Li, Zhiyuan Zhang, and Junchi Yan. 2024. Bench2Drive: Towards Multi-Ability Benchmarking of Closed-Loop End-To-End Autonomous Driving. arXiv:2406.03877.
 - [25] Feng Jiang, Yang Chen, Kyle Xu, Yuchen Liu, Haifeng Wang, Zhenhao Shen, Jasper Lu, Shengze Huang, Yuanfei Wang, Chen Xie, and Ruihai Wu. 2026. RoboWM-Bench: A Benchmark for Evaluating World Models in Robotic Manipulation. arXiv:2604.19092.
 - [26] Mukul Khanna, Ram Ramrakhya, Gunjan Chhablani, Sriram Yenamandra, Theophile Gervet, Matthew Chang, Zsolt Kira, Devendra Singh Chaplot, Dhruv Batra, and Roozbeh Mottaghi. 2024. GOAT-Bench: A Benchmark for Multi-Modal Lifelong Navigation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. arXiv:2404.06609.
 - [27] Vikash Kumar, Rutav Shah, Gaoyue Zhou, Vincent Moens, Vittorio Caggiano, Jay Vakil, Abhishek Gupta, and Aravind Rajeswaran. 2023. RoboHive: A Unified Framework for Robot Learning. arXiv:2310.06828.
 - [28] Jinshan Lai, Jianwei Hu, Baoyang Jiang, Fengchun Zhang, Leyuan Wang, Haotian Li, Yida Wang, Tingxuan Huang, Xi Ren, and Qiang Ma. 2026. Intelligent Automation for Embodied Benchmark Construction: Pipelines, Embodiments, Simulators, and Trends. arXiv:2606.12207.
 - [29] Chengshu Li, Ruohan Zhang, Josiah Wong, Cem Gokmen, Sanjana Srivastava, Roberto Martin-Martin, Chen Wang, Gabrael Levine, Wensi Ai, Benjamin Martinez, Hang Yin, Michael Lingelbach, Minjune Hwang, Ayano Hiranaka, Sujay Garlanka, Arman Aydin, Sharon Lee, Jiankai Sun, Mona Anvari, Manasi Sharma, Dhruva Bansal, Samuel Hunter, Kyu-Young Kim, Alan Lou, Caleb R Matthews, Ivan Villa-Renteria, Jerry Huayang Tang, Claire Tang, Fei Xia, Yunzhu Li, Silvio Savarese, Hyowon Gweon, C. Karen Liu, Jiajun Wu, and Li Fei-Fei. 2024. BEHAVIOR-1K: A Human-Centered, Embodied AI Benchmark with 1,000 Everyday Activities and Realistic Simulation. arXiv:2403.09227.
 - [30] Quanyi Li, Zhenghao Peng, Lan Feng, Qihang Zhang, Zhenghai Xue, and Bolei Zhou. 2021. MetaDrive: Composing Diverse Driving Scenarios for Generalizable Reinforcement Learning. arXiv:2109.12674.
 - [31] Xuanlin Li, Kyle Hsu, Jiayuan Gu, Karl Pertsch, Oier Mees, Homer Rich Walke, Chuyuan Fu, Ishikaa Lunawat, Isabel Sieh, Sean Kirmani, Sergey Levine, Jiajun Wu, Chelsea Finn, Hao Su, Quan Vuong, and Ted Xiao. 2024. Evaluating Real-World Robot Manipulation Policies in Simulation. arXiv:2405.05941.
 - [32] Mingxiang Liao, Hannan Lu, Xinyu Zhang, Fang Wan, Tianyu Wang, Yuzhong Zhao, Wangmeng Zuo, Qixiang Ye, and Jingdong Wang. 2024. Evaluation of Text-to-Video Generation Models: A Dynamics Perspective. arXiv:2407.01094.
 - [33] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. 2023. LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning. arXiv:2306.03310.
 - [34] Yaofang Liu, Xiaodong Cun, Xuebo Liu, Xintao Wang, Yong Zhang, Haoxin Chen, Yang Liu, Tiejong Zeng, Raymond Chan, and Ying Shan. 2023. EvalCrafter: Benchmarking and Evaluating Large Video Generation Models. arXiv:2310.11440.
 - [35] Jianlan Luo, Charles Xu, Fangchen Liu, Liam Tan, Zipeng Lin, Jeffrey Wu, Pieter Abbeel, and Sergey Levine. 2024. FMB: a Functional Manipulation Benchmark for Generalizable Robotic Learning. arXiv:2401.08553.
 - [36] Nicolai A. Lynnerup, Laura Nolling, Rasmus Hasle, and John Hallam. 2020. A Survey on Reproducibility by Evaluating Deep Reinforcement Learning Algorithms on Real-World Robots. In *Conference on Robot Learning (CoRL)*, PMLR vol. 100. arXiv:1909.03772.
 - [37] Arjun Majumdar, Karmesh Yadav, Sergio Arnaud, Yecheng Jason Ma, Claire Chen, Sneha Silwal, Aryan Jain, Vincent-Pierre Berges, Pieter Abbeel, Jitendra Malik, Dhruv Batra, Yixin Lin, Oleksandr Maksymets, Aravind Rajeswaran, and Franziska Meier. 2023. Where are we in the search for an Artificial Visual Cortex for Embodied Intelligence? arXiv:2303.18240.
 - [38] Roberto Martin-Martin, Mihir Patel, Hamid Rezafofighi, Abhijeet Sheno, JunYoung Gwak, Eric Frankel, Amir Sadeghian, and Silvio Savarese. 2019. JRDB: A Dataset and Benchmark of Egocentric Robot Visual Perception of Humans in Built Environments. arXiv:1910.11792.
 - [39] Matthew J. Page, Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, et al. 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 372 (2021), n71. doi:10.1136/bmj.n71.
 - [40] Noe Perez-Higueras, Roberto Otero, Fernando Caballero, and Luis Merino. 2023. HuNavSim: A ROS 2 Human Navigation Simulator for Benchmarking Human-Aware Robot Navigation. arXiv:2305.01303.
 - [41] Xavier Puig, Eric Undersander, Andrew Szot, Mikael Dallaire Cote, Tsung-Yen Yang, Ruslan Partsey, Ruta Desai, Alexander William Clegg, Michal Hlavac, So Yeon Min, Vladimir Vondrus, Theophile Gervet, Vincent-Pierre Berges, John M. Turner, Oleksandr Maksymets, Zsolt Kira, Mrinal Kalakrishnan, Jitendra Malik, Devendra Singh Chaplot, Unnat Jain, Dhruv Batra, Akshara Rai, and Roozbeh Mottaghi. 2023. Habitat 3.0: A Co-Habitat for Humans, Avatars and Robots. arXiv:2310.13724.
 - [42] Wilbert Pumacay, Ishika Singh, Jiafei Duan, Ranjay Krishna, Jesse Thomason, and Dieter Fox. 2024. THE COLOSSEUM: A Benchmark for Evaluating Generalization for Robotic Manipulation. arXiv:2402.08191.

- [43] Shi Qiu, Shaoyang Guo, Zhuo-Yang Song, Yunbo Sun, Zeyu Cai, Jiashen Wei, Tianyu Luo, and Yixuan Yin. 2025. PHYBench: Holistic Evaluation of Physical Perception and Reasoning in Large Language Models. [arXiv:2504.16074](#).
- [44] Santhosh K. Ramakrishnan, Aaron Gokaslan, Erik Wijmans, Oleksandr Maksymets, Alex Clegg, John Turner, Eric Undersander, Wojciech Galuba, Andrew Westbury, Angel X. Chang, Manolis Savva, Yili Zhao, and Dhruv Batra. 2021. Habitat-Matterport 3D Dataset (HM3D): 1000 Large-scale 3D Environments for Embodied AI. [arXiv:2109.08238](#).
- [45] Ronan Riochet, Mario Yncente Castro, Mathieu Bernard, Adam Lerer, Rob Fergus, Veronique Izard, and Emmanuel Dupoux. 2018. IntPhys: A Framework and Benchmark for Visual Intuitive Physics Reasoning. [arXiv:1803.07616](#).
- [46] Manolis Savva, Abhishek Kadian, Oleksandr Maksymets, Yili Zhao, Erik Wijmans, Bhavana Jain, Julian Straub, Jia Liu, Vladlen Koltun, Jitendra Malik, Devi Parikh, and Dhruv Batra. 2019. Habitat: A Platform for Embodied AI Research. [arXiv:1904.01201](#).
- [47] Yu Shang, Zhuohang Li, Yiding Ma, Weikang Su, Xin Jin, Ziyu Wang, Lei Jin, Xin Zhang, Yinzhou Tang, Haisheng Su, Chen Gao, Wei Wu, Xihui Liu, Dhruv Shah, Zhaoxiang Zhang, Zhibo Chen, Jun Zhu, Yonghong Tian, Wenwu Zhu, and Yong Li. 2026. WorldArena: A Unified Benchmark for Evaluating Perception and Functional Utility of Embodied World Models. [arXiv:2602.08971](#).
- [48] Yu Shang, Yinzhou Tang, Xin Zhang, Shengyuan Wang, Yuwei Yan, Honglin Zhang, Zhiheng Zheng, Jie Zhao, Jie Feng, Chen Gao, Jiansheng Chen, and Yong Li. 2026. A Survey of Embodied World Models. Preprint (non-arXiv).
- [49] Mohit Shridhar, Jesse Thomason, Daniel Gordon, Yonatan Bisk, Winson Han, Roozbeh Mottaghi, Luke Zettlemoyer, and Dieter Fox. 2019. ALFRED: A Benchmark for Interpreting Grounded Instructions for Everyday Tasks. [arXiv:1912.01734](#).
- [50] Andrew Szot, Alex Clegg, Eric Undersander, Erik Wijmans, Yili Zhao, John Turner, Noah Maestre, Mustafa Mukadam, Devendra Chaplot, Oleksandr Maksymets, Aaron Gokaslan, Vladimir Vondrus, Sameer Dharur, Franziska Meier, Wojciech Galuba, Angel Chang, Zsolt Kira, Vladlen Koltun, Jitendra Malik, Manolis Savva, and Dhruv Batra. 2021. Habitat 2.0: Training Home Assistants to Rearrange their Habitat. [arXiv:2106.14405](#).
- [51] Ziyao Wang, Bingying Wang, Hanrong Zhang, Tingting Du, Tianyang Chen, Guoheng Sun, Yexiao He, Zheyu Shen, Wanghao Ye, and Ang Li. 2026. Vision-Language-Action in Robotics: A Survey of Datasets, Benchmarks, and Data Engines. [arXiv:2604.23001](#).
- [52] Fei Xia, Amir Zamir, Zhi-Yang He, Alexander Sax, Jitendra Malik, and Silvio Savarese. 2018. Gibson Env: Real-World Perception for Embodied Agents. [arXiv:1808.10654](#).
- [53] Xuemeng Yang, Licheng Wen, Yukai Ma, Jianbiao Mei, Xin Li, Tiantian Wei, Wenjie Lei, Daocheng Fu, Pinlong Cai, Min Dou, Botian Shi, Liang He, Yong Liu, and Yu Qiao. 2024. DriveArena: A Closed-loop Generative Simulation Platform for Autonomous Driving. [arXiv:2408.00415](#).
- [54] Kexin Yi, Chuang Gan, Yunzhu Li, Pushmeet Kohli, Jiajun Wu, Antonio Torralba, and Joshua B. Tenenbaum. 2019. CLEVRER: CoLLision Events for Video REpresentation and Reasoning. [arXiv:1910.01442](#).
- [55] Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Avnish Narayan, Hayden Shively, Adithya Bellathur, Karol Hausman, Chelsea Finn, and Sergey Levine. 2019. Meta-World: A Benchmark and Evaluation for Multi-Task and Meta Reinforcement Learning. [arXiv:1910.10897](#).
- [56] Yang Yu, Shiyuan Zhang, Yifei Sheng, Haoxiang Ren, and Haoxin Lin. 2026. How Should World Models Be Evaluated for Embodied Decision-Making? A Decision-Making-Centric Position. [arXiv:2606.15032](#).
- [57] Hongxin Zhang, Weihua Du, Jiaming Shan, Qinhong Zhou, Yilun Du, Joshua B. Tenenbaum, Tianmin Shu, and Chuang Gan. 2023. Building Cooperative Embodied Agents Modularly with Large Language Models. [arXiv:2307.02485](#).
- [58] Jiahan Zhang, Muqing Jiang, Nanru Dai, Taiming Lu, Arda Uzunoglu, Shunchi Zhang, Yana Wei, Jiahao Wang, Vishal M. Patel, Paul Pu Liang, Daniel Khashabi, Cheng Peng, Rama Chellappa, Tianmin Shu, Alan Yuille, Yilun Du, and Jieneng Chen. 2025. World-in-World: World Models in a Closed-Loop World. [arXiv:2510.18135](#).
- [59] Zhicheng Zheng, Xin Yan, Zhenfang Chen, Jingzhou Wang, Qin Zhi Eddie Lim, Joshua B. Tenenbaum, and Chuang Gan. 2024. ContPhy: Continuum Physical Concept Learning and Reasoning from Videos. [arXiv:2402.06119](#).